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College of Informatics and Computing Sciences

PROJECT PROPOSAL
Computer Science Department
2nd Semester AY 2025-2026

Course Code	CC 102
Course Description	Advanced Computer Programming
Section	CS 1205
Date	March 08, 2026
Time	

SR-Code	25-03188
Name of Student	Babao, Althea Clarine L.



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MEDIQUEUE: SMART HOSPITAL QUEUE SYSTEM

I. Problem Description and Analysis

General Problem Description

Traditional hospital queuing often involves physical lines and manual ticket systems. This leads to overcrowded waiting rooms, unorganized queue, increased risk of cross-infection, and create stress due to “waiting blindness”, where patients have no real-time data on their expected waiting time.

Specific Sub-Problems

From the general problem, the following specific sub-problems are identified:

- 1. Waiting Room Overcrowding**
Patients must remain physically present, leading to congested corridors.
- 2. Lack of Real-Time Updates**
Patients are unaware of their position in the queue or sudden delays caused by emergencies.
- 3. Unmonitored Cycle**
Hospital staff struggle to manually prioritize patients based on symptom severity while managing a high volume of entries.

Intended System Users

1. Patients – they use the system to register their personal information, monitor their position in the queue, and view the estimated waiting time through the GUI.
2. Staff (Receptionists, Nurses, Doctors) – they use the system to manage the queue, call patients, mark patients, prioritize emergencies, and monitor overall patient flow.

Suitability for a Python-Based Solution

Python is ideal for this MediQueue because its libraries (Tkinter or PyQt) allow for rapid GUI development. Its robust data handling capabilities enable real-time calculations, while its modularity allows for easy integrations of notification simulations and data logging.

II. System Objectives and Requirements

Main Objective

To develop a Python-based GUI application that digitizes the queue process, provides the patient real-time estimations, and streamlines patient flow

Functional Requirements

The proposed system shall:

1. Patient Registration allows patients to input name, contact info, and symptom severity.
2. Displays a real-time list of current queue numbers and estimated waiting times
3. Enable staff to “call” the next patient, mark a patient re-prioritize emergency cases.

Expected Program Inputs and Outputs

Inputs: Patient Name, ID, Symptom type (e.g., General, Urgent, Emergency), Staff Menu Selection, and Check-in Time.

Outputs: Digital Queue Number, ETA, Queue Dashboard Display, Patient Status Update, and System Messages.

III. Proposed Data Structures

The system will utilize the following Python data structures:

Datetime module: to record patient check-in times and calculate Estimated Waiting Time (ETA).



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Dictionary: to store individual patient records with details like name, ID, severity, symptoms, queue number and check-in time

List: to record and store multiple pieces of data of the user, and to manage the queue order of patients

Justification

Using datetime ensures accurate time tracking for dynamic ETA calculation. Dictionaries allow ensure access and updates of patient information without affecting other records. List preserve the queue order and make it simple to implement priority handling, such as moving emergency cases to the front.

IV. Control Structures and Program Flow

Use of Control Structures

- **Conditions:**
If-elif-else statements will categorize patients by severity.
For example:
If “Symptom == Emergency”, the patient is moved to the front of the list.
- **Loops:**
While loops will keep the GUI active, and for loops will be used to recalculate the ETA for every person in the list whenever a new patient is added.
- **Recursion:**
A recursion function can be used to reorder queue for emergencies dynamically

Program Flow Description

1. The program will ask for the username.
2. The program will show the menu options.
3. The user will input the data based on the chosen option.
4. The program will store and calculate the data based on the user’s choice.
5. The program will perform the needed actions/calculations.
6. The program will display or save the results.
7. The program will send notifications to the user.

Key Decision Points

Following the program, the system checks if the patient input is valid, determines the patient’s priority, verifies if the queue is empty, and processes the staff’s menu selection, then takes the appropriate action such as adding the patient to the queue, updating the dashboard, calling the next patient, or prioritizing an emergency case.



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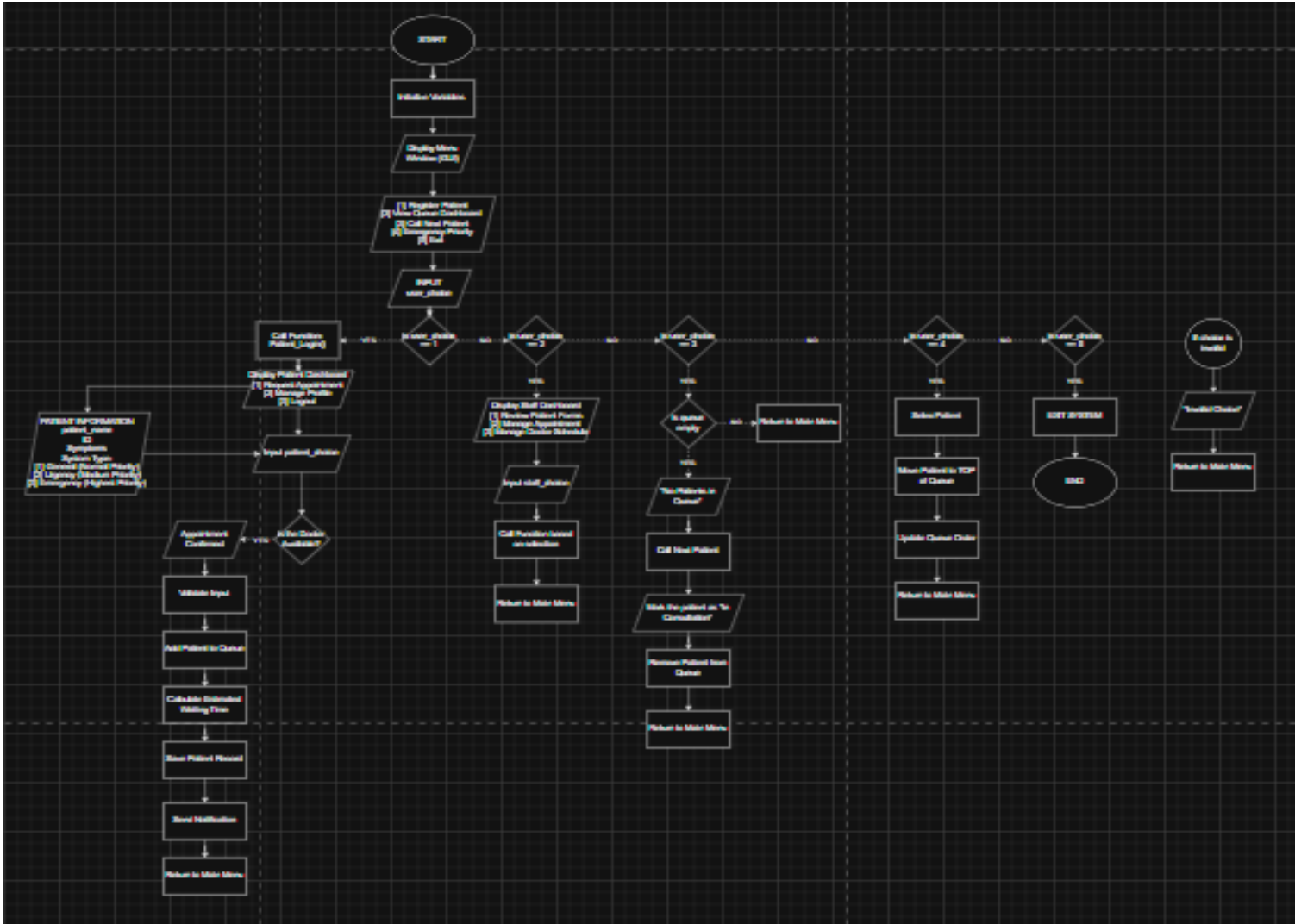
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V. High-Level Solution Design (Program Outline) this will become 4th
Proposed Program Structure (Modular Design)

FLOWCHART:





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QUEUE FLOW DIAGRAM:

